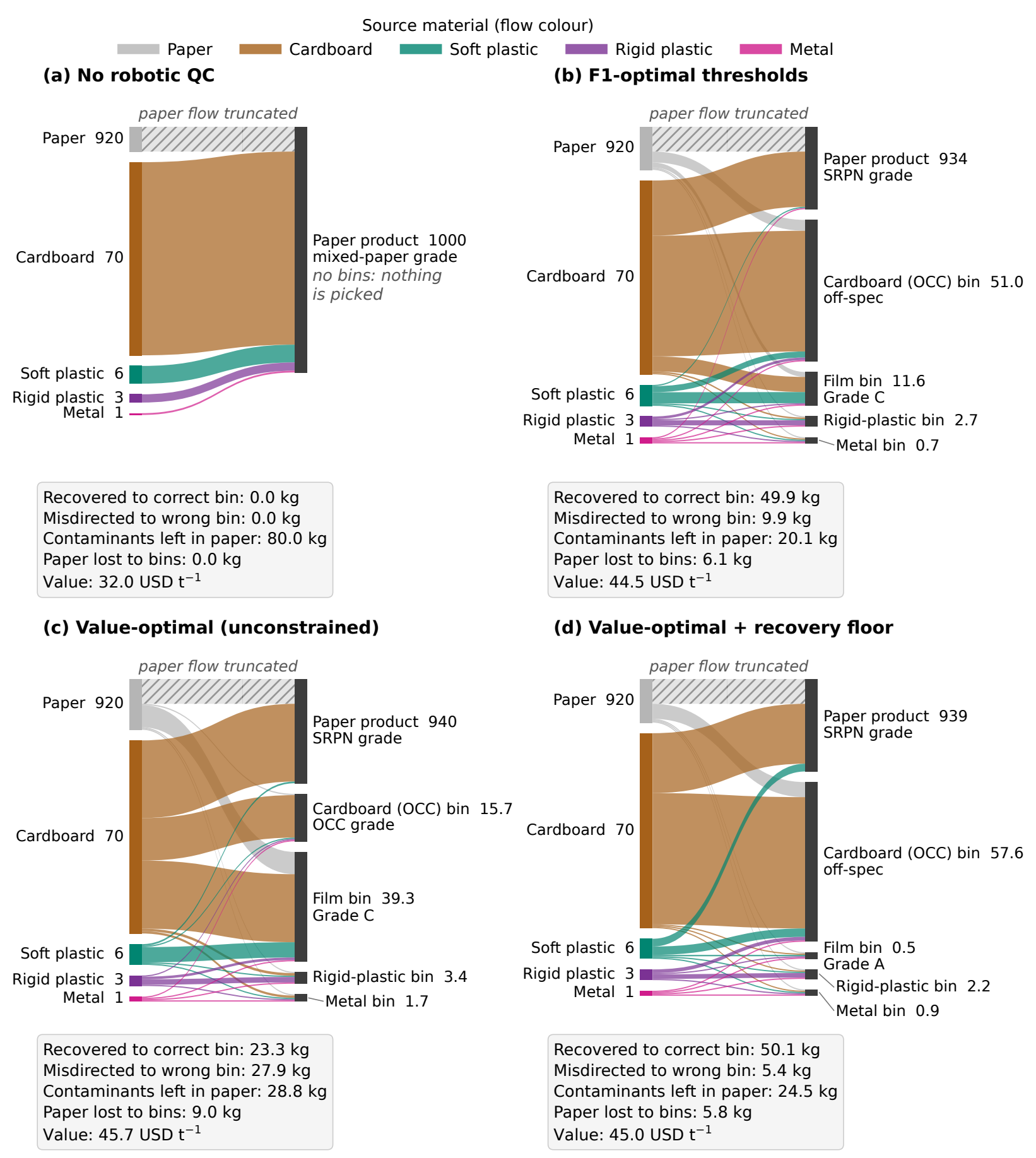
Source material (flow colour)
Paper
Cardboard
Soft plastic
Rigid plastic
Metal
(a) No robotic QC
paper flow truncated
Paper 920
Cardboard 70
Soft plastic 6
Rigid plastic 3
Metal 1
Paper product 1000
mixed-paper grade
no bins: nothing
is picked
Recovered to correct bin: 0.0 kg
Misdirected to wrong bin: 0.0 kg
Contaminants left in paper: 80.0 kg
Paper lost to bins: 0.0 kg
Value: 32.0 USD t−1
(b) F1-optimal thresholds
paper flow truncated
Paper 920
Cardboard 70
Soft plastic 6
Rigid plastic 3
Metal 1
Paper product 934
SRPN grade
Cardboard (OCC) bin 51.0
off-spec
Film bin 11.6
Grade C
Rigid-plastic bin 2.7
Metal bin 0.7
Recovered to correct bin: 49.9 kg
Misdirected to wrong bin: 9.9 kg
Contaminants left in paper: 20.1 kg
Paper lost to bins: 6.1 kg
Value: 44.5 USD t−1
(c) Value-optimal (unconstrained)
paper flow truncated
Paper 920
Cardboard 70
Soft plastic 6
Rigid plastic 3
Metal 1
Paper product 940
SRPN grade
Cardboard (OCC) bin 15.7
OCC grade
Film bin 39.3
Grade C
Rigid-plastic bin 3.4
Metal bin 1.7
Recovered to correct bin: 23.3 kg
Misdirected to wrong bin: 27.9 kg
Contaminants left in paper: 28.8 kg
Paper lost to bins: 9.0 kg
Value: 45.7 USD t−1
(d) Value-optimal + recovery floor
paper flow truncated
Paper 920
Cardboard 70
Soft plastic 6
Rigid plastic 3
Metal 1
Paper product 939
SRPN grade
Cardboard (OCC) bin 57.6
off-spec
Film bin 0.5
Grade A
Rigid-plastic bin 2.2
Metal bin 0.9
Recovered to correct bin: 50.1 kg
Misdirected to wrong bin: 5.4 kg
Contaminants left in paper: 24.5 kg
Paper lost to bins: 5.8 kg
Value: 45.0 USD t−1
All values in kg per tonne of feed. Flow widths share one scale, except the truncated (hatched) paper flow.